

Annual Report

2018-19



Indian Institute of Information Technology
Pune

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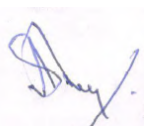
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1. Director's Message



Congratulations on your academic achievements, Indian Institute of Information Technology, Pune (IIIT Pune) established in 2016, is one among the fastest growing IIITs in 20 newly established IIITs by the Ministry of Human Resource Development (MHRD), Government of India and industry partners as Not-for-profit Public Private Partnership (N-PPP) institution. ROLTA Ltd. and HUBTOWN Ltd. are the industry partners of IIIT Pune. IIIT Pune is dedicated towards imparting high quality technical education and a long term vision to the students. Our focus lies not just on academic brilliance but also on knowledge that is pervasive in areas of emerging technologies. We, through our learning and research programs aim at building such applications of information technology whose benefits percolate even to grass root levels. We, not only believe in making our students the superior scholars but responsible members of society. Ours is a complete institute in the truest sense of the term, endowed with facilities and an environment that is conducive to learning. The institute focuses on interaction with the best institutes in terms of academic exchange programmes, transfer of credits, internships, industry-academia meets, etc. Every department in the institute has state-of-the-art laboratories. Several Industry-Institute partnership programmes, soft-skill trainings, competitions, internships and student projects are organized to provide communication skills, commitment, and ethics for comprehensive growth of an individual. The institute promotes innovation and entrepreneurship by encouraging and supporting innovative and out-of-the-box ideas. Transformation doesn't just happen. It takes a plan and a support system. So, while studying at IIIT Pune, you'll be expected to work hard, have a true and right attitude in the subject area. Walk boldly and confidently through the next leg of your life's exciting and fruitful journey.

Best of luck!

A handwritten signature in blue ink, appearing to read 'Anupam Shukla', written over a light blue grid background.

Prof. Anupam Shukla
Director, IIIT Pune

2. Registrar's Message



I am glad to welcome you to Indian Institute of Information Technology, Pune (IIIT Pune). It was established in 2016 and is an Institute of National importance under the Act of Parliament. IIIT Pune endeavors to be at par with the international standards, to provide a unique learning experience to the students, and aspires to be a world class centre of excellence in teaching, research and innovation in cutting-edge technologies. The institute promotes innovative and entrepreneurial thinking by the students and intends to provide help to them to achieve such mindset. Our mission is to achieve academic excellence in engineering, technology and science by imparting quality and value based education. We are working hard to inspire our students to become responsible citizens and competent professionals with high ethical values. We are trying to satisfy the academic as well as infrastructure facilities of the students who are coming from various corners of the country. With the able administrative vision of Dr. Anupam Shukla, Director, IIIT Pune, it is venturing towards achieving international standards in all respect. We are also trying to create infrastructure facilities in a permanent campus at the earliest possible at Talegaon, near old Pune-Mumbai highway. I am very much thankful to all the students and faculty for supporting the institute in many respects.

Wishing the faculty and students all the very best!

A handwritten signature in blue ink, appearing to read 'S.N. Sapali'.

Dr. S.N. Sapali
I/C Registrar, IIIT Pune

3. Preface

Indian Institute of Information Technology, Pune (IIIT Pune) is an institute of national importance set up under the purview of the Indian Institute of Information Technology public private partnership (PPP) Act 2017, with a view to develop new knowledge in information technology and to provide manpower of global standards for the information technology industry. IIIT Pune was established by MHRD, Govt. of India on not for profit basis in PPP with Government of Maharashtra and industry partners. IIIT Pune is registered as society under Maharashtra Society Registration Act 1860 on January 07th, 2016. IIIT Pune is a joint venture of Govt. of India (50%), Govt. of Maharashtra (35%) and Industrial Partners (15%). The institute started its first academic session from 1st August 2016 with intake in two disciplines namely, Electronics and Communication Engineering (ECE), Computer Science and Engineering (CSE) with intake capacity of 60 students each.

This is the third annual report of IIIT Pune, brought out to provide a glimpse of its performance and achievements during the year 2018-19. This report gives a bird's eye view of all the major activities carried out during this year.

3.1 Vision

To be an institution of eminence globally recognized for its academic excellence through research and innovation, industry engagement evolving professional culture.

3.2 Mission

- To nurture industry practices and to collaborate with industry towards technology development.
- To focus on student development, cultivating skills and innovation that builds problem-solving skills.
- To foster a culture of innovation and research among the students and faculties.
- To associate with ranked (National and Global) institutes for Collaborative engagement
- To be the preferred institute for knowledge seekers.

3.3 Objectives

- To achieve international standards in student-faculty ratio, doctoral students to postgraduate student's ratio, research output, publications in quality journals and placement of students.
- To achieve growth by increasing the number of participants in its program of Post graduate programme, Education development programme, and Fellowship.
- To maintain a high level of student satisfaction with the courses being taught as well as with the placements.
- As part of the social responsibility, to actively participate in various social activities, flagship programmes and missions being undertaken.
- To improve the socio-economic environment in and around the institute so as to have an impact on the quality of life of people.

4. Administration

IIIT Pune is an institute of national importance and is governed by the IIIT (PPP) Act 2017. Responsibility for the general superintendence, direction and control of the affairs of the institute is vested in the Board of Governors (BoG). The board functions through its standing committees including the Senate, the Finance Committee, and the Buildings and Works Committee (BWC) and other such committees which may be constituted to consider specific issues.

4.1 Board of Governors

The composition of the BoG is as follows:

1	Shri. R. Subrahmanyam	Chairman	Secretary, Technical Education, MHRD, New Delhi
2	Dr. S. S. Sandhu	Member	Additional Secretary, Technical Education, MHRD, New Delhi
3	Ms. Darshana M. Dabral	Member	Director, Finance, MHRD, New Delhi
4	Shri. Prashant Agarwal	Member	Director (IITs and IIITs), MHRD, New Delhi
5	Dr. Pramod M. Padole	Member	Director, Visvesvaraya National Institute of Technology (VNIT), Nagpur
6	Dr. D R Nandanwar	Member	Joint Director, Technical Education, Pune Region, Pune
7	Dr. Anupam Shukla	Member	Director, IIIT Pune
8	Dr. S. N. Sapali	Invitee	I/C Registrar, IIIT Pune
9	Shri. Praveer Saxena	Invitee	Under Secretary, MHRD, Govt. of India

4.2 Senate

The composition of the Senate is as follows:

1	Dr. Anupam Shukla	Chairman	Director, IIIT Pune
2	Prof. Laxmidhar Behera	Member	Professor, IIT, Kanpur
3	Mr. Anuj Mehra	Member	Lead Data scientist, Magic Bricks, Bangalore
4	Ms. Yesoda Bhargava	Member	Sr. Lead Engineer, Samsung Research, Bangalore
5	Prof. Jagannath Aghav	Member	Professor, CoE, Pune
6	Dr. Shekhar Bhansali	Member	Professor, Florida International University, USA
7	Dr. Girijesh Prasad	Member	Ulster University, UK
8	Dr. Debasish Ghose	Member	IISc, Bangalore
9	Dr. S. N. Sapali	Secretary	I/C Registrar, IIIT Pune

4.3 Building and Works Committee

The composition of the BWC is as follows:

1	Dr. Anupam Shukla	Chairman	Director, IIIT Pune
2	Prof K K Pathak	Member	Dept. of Civil Engineering IIT (BHU) Varanasi – 221005
3	Mr Suresh Laxman Ganeshkar	Member	Chief Engineer MSEDCL, Vidyut Bhawan, Old Power House Compound, Mill Corner, Aurangabad-431001
4	Prof. Sanjeev Singh	Member	Head, Dept. of Architecture SPA Neelbad Road, Bhauri, Bhopal-462030
5	Shri B G Birajdar	Member	Dept. of Civil Engineering, CoE Shivajinagar, Pune-411005
6	Shri Chandrakant P Jawale	Member	SE, PMRDA 3 rd Floor, PCNTDA New Administrative Building, Akurdi, Pune-411044
7	Shri Anil Kumar Jain	Member	Former DG, CPWD, Plot F-21-C, Sector-50 Noida

8	Shri S Ramanujam	Member	Member, BWC (IITD/IIT Ropar / IIT Tirupati)
9	Shri Niranjana Singh	Member	Former CE, CPWD, A-3/201, Nirmal Chhaya Tower VIP Road, SAS Nagar, Mohali (Punjab)
10	Prof. Neeraj Gupta	Member	Dept. of Architecture Central University of Rajasthan, 424, Sector 7, Malviya Nagar Jaipur-302017
11	Prof. Rajesh K Tripathi	Member	Dept. of Civil Engineering, NIT Raipur (C.G)

4.4 Administrative Activities

- 3rd Meeting of BoG of IIIT Pune was held on February 26th, 2019 at MHRD, New Delhi.
- **MoU with premier Institutes of India:** IIIT Pune has signed the memorandum of understanding for research and academic collaborations with premiere institutes including DIAT Pune, IIT-BHU, NIT Jaipur, NDA Pune. The students and faculties of IIIT Pune can access various facilities of the institute and start collaborative research and training programs.
- **Setting up of National Resource Center:** IIIT Pune has set up a National Resource Centre for CSE and ECE for advance refresher program in teaching under ‘Pandit Madan Mohan Malaviya National Mission on Teachers and Teaching’ (PMMNMTT) scheme of MHRD.
- **MoU with Central Public Works Department (CPWD):** IIIT Pune has signed MoU with CPWD for construction of boundary wall and the permanent campus.
- **Recruitment of contractual faculties:** Recruitment of contractual faculties, in the department of ECE and CSE was carried out in June 2018 and Dec 2018.
- **Government e-Market Place (GeM) Implementation:** IIIT Pune has successfully implemented GeM for making the purchasing of various products for the institute easier.

5. Academic

5.1 Department and Academic Programmes

IIIT Pune is committed to provide the best in class technical education in the country. The institute is facilitating knowledge transfer whilst keeping up with the latest development pedagogy. The institute offers four-year undergraduate [Bachelor of Technology (B.Tech.)] and Doctor of Philosophy (Ph.D.) programs in following departments

- a) Electronics and Communication Engineering (ECE)
- b) Computer Science and Engineering (CSE).

5.2 Intake of the year (Admission Status)

The admissions for B.Tech in the institute are carried out by Central Seat Allocation Board (CSAB) by the Joint Seat Allocation Authority (JoSAA).

Programme and specialization	Intake
B.Tech. (CSE)	60
B.Tech. (ECE)	60

The Ph.D. programme is aimed at assisting students in acquiring proficiency in their chosen area of research. The academic programme leading to the Ph.D. degree is broad-based and involves a course credit requirement and a research thesis submission. The institute also encourages research in interdisciplinary areas through a system of joint supervision and inter-departmental group activities. Research activities in IIIT Pune are carried out under the ambit of academic departments (ECE and CSE) and research centres (a) Centre of Robotics and Security in Internet of Things (IoT) Space, (b) Centre of Indian Languages and Computational Intelligence, (c) Centre of VLSI and Nanotechnology. With an aim to develop new knowledge in all aspects of teaching and research, IIIT Pune invites applications from eligible Indian nationals for the admission to the Ph.D. program of the institute. IIIT Pune plans to develop world-class research facilities in areas of Software Engineering, Mobile Computing, Information Security, IoT, Robotics, Machine Learning, Speech and Image Processing, Medical Informatics, Cyber Physical Systems, Wireless Networks, VLSI Design and Nanotechnology.

Number of students admitted into various programs of the institute in 2018-19.

Sr. No.	Class	Boys	Girls	Total
1	B.Tech ECE	31	6	37
2	B.Tech CSE	44	2	46

IIIT Pune is actively working on revamping its UG curriculum to meet the future IT requirements. Apart from academics, the institute provides comprehensive environment that promotes art, culture, sports, societal contributions and self-governance. UG students participate in ongoing research and technology development, as a result, a vibrant UG program exists to fulfill the needs of the society.

5.3 List of Faculty Members

S. No.	Name of Faculty	Designation
1	Dr. Sourabh Shrivastava	Assistant Professor, CSE
2	Dr. Bhupendra Singh	Assistant Professor, CSE
3	Dr. Tanmoy Hazra	Assistant Professor, CSE
4	Dr. Renu Kachhoria	Assistant Professor, CSE
5	Dr. Darshan Medhane	Assistant Professor, CSE
6	Dr. Jaideep Pati	Assistant Professor, CSE
7	Dr. Pragati Agrawal	Assistant Professor, CSE
8	Dr. Saif Nalband	Assistant Professor, CSE
9	Dr. Sanjeev Sharma	Assistant Professor, CSE
10	Dr. Nagendra Kushwaha	Assistant Professor, ECE
11	Dr. Parul Sahare	Assistant Professor, ECE
12	Dr. Sandeep Mishra	Assistant Professor, ECE
13	Dr. Pankaj Sharma	Assistant Professor, ECE
14	Dr. Mayur Parate	Assistant Professor, ECE
15	Dr. Sandeep Kumar Singh	Assistant Professor, ECE
16	Dr. Ranjith R. Nair	Assistant Professor, ECE
17	Dr. Priyabrat Garnaik	Assistant Professor, ECE
18	Dr. Dheeraj Dhane	Assistant Professor, ECE
19	Dr. Chetan Khadse	Assistant Professor, ECE

20	Dr. Anagha Khiste	Assistant Professor, Mathematics
21	Dr. Chandrakant Guled	Assistant Professor, Mathematics
22	Dr. Meena More	Assistant Professor, Mathematics
23	Dr. Arjun Tarale	Assistant Professor, Physics
24	Dr. Satish Mondal	Assistant Professor, Physics
25	Dr. Nilofar Nadaf	Assistant Professor, Biology
26	Ms. Rukhsheen Vajifdar	Adjunct Faculty, English

5.4 Examination Results

Top three rank holders in each of the branches of Engineering after the Examination held in December 2018:

Sr. No	Branch	Year	Name of the students	MIS	CGPA	Rank
1	CSE	First	Aman Singhal	111815005	9.55	1
			Preet Bhatia	111815027	9.51	2
			Pankaj Kumar Gupta	111815025	9.47	3
		Second	KNS Kiran Sai Karra	111715028	9.44	1
			Abhishek Nitin Jugdar	111715003	9.17	2
			Shabbir Chatrissa	111715041	9.04	3
		Third	Dudhal Disha	111615011	9.34	1
			Yogesh Karan	111615036	9.2	2
			Sachin Kanth	111616023	9.05	3
2	ECE	First	Devang Bhagwat	111816011	9.21	1
			Mayank Goel	111816023	8.94	2
			Shreyansh Gupta	111816034	8.62	3
		Second	Rutuja Madhure	111716029	9.93	1
			Radha Raman	111716026	9.46	2
			Karan Khanna	111716018	9.29	3
		Third	Saiphanish Todupunuri	111616025	8.92	1
			Anujkumar Bhaskar	111616004	8.83	2
			Jidnyesha Patil	111616021	8.29	3

5.5 Facilities

5.5.1 Various Laboratories at IIIT Pune

1. Artificial Intelligence and Robotics lab
2. Python Programming and Machine learning lab
3. VLSI Design and Nanotechnology lab
4. Wireless communication lab
5. RF and Microwave lab
6. Electronic Devices lab
7. Advance programming lab

5.5.2 Setting up of National Resource Center

IIIT Pune is a national resource center for CSE and ECE. IIIT Pune has already developed two lecture series by Dr. Joydip Dhar, ABV-IIITM, Gwalior and Dr. Laxmidhar Behera, IIT, Kanpur.

5.5.3 Library and Information Center

The library at IIIT Pune has been serving the students with the required books and other learning material. Library is having 332 book titles and 3632 volumes, which is gradually increasing every semester. The library has 1335 books belonging to ECE, 1183 books pertaining to CSE, and 1114 books for first year students. It also has different national and international journals. The current annual budget for library is Rs 6.6 Lakhs.

6. Research and Development

6.1 Research Centers

IIIT Pune has established research centres in following identified areas

1. Centre for Robotics and Security in IoT Space.
2. Centre for VLSI and Nanotechnology.
3. Centre for Indian Languages and Computational Intelligence.

The faculty members of IIIT Pune are associated with various research centres for multidisciplinary research. Different activities have been started by the research centers including conducting Faculty Development Program (FDPs) and Short Term Training Programs (STP), collaborative research, etc.

6.2 Activities under Research Centres

6.2.1 Organizing STP

Centre for Robotics and IoT, IIIT Pune organized a STP on Robotics and IoT for students, faculties and industry persons from 8th–12th December, 2018. People from various states of India participated in the event. The programme included various types of sessions, theoretical, hands on and industrial demonstration. The response from the participants was tremendous. Dr. Sandeep Kumar Singh and Dr. Nagendra Kushwaha were the coordinators of this event.

6.2.2 Organizing Faculty Development Programme (FDP)

Centre for Indian Languages and Computational Intelligence, IIIT Pune organized a five-day FDP on Data Science and Machine Learning for faculties and industry persons from 20th–24th December, 2018. Fifteen participants from various states of India participated in the event. The response from the participants was tremendous. The dignitaries present during the inauguration ceremony were Prof. Anupam Shukla, Director IIIT Pune, Prof. Debasish Ghose, Professor IISc, Bangalore and Prof. U. Ghanekar, Professor, NIT Kurukshetra. Dr. Tanmoy Hazra and Dr. Ranjith Nair were coordinators of the event.



A glimpse of FDP Sessions

6.3 Faculty Publications

S. No.	Details	Pub. Year	Indexed
1	T. Hazra, C. R. S. Kumar, M. Nene, Two-agent cooperative search model using Petri-nets, <i>IJIUS</i> , 6, 1-12	2018	Scopus
2	S. Mishra, T. V. Mahendra, J. Saikia, and A. Dandapat, “A low-overhead dynamic TCAM with pipelined read-restore refresh scheme,” <i>IEEE Transactions on Circuits and Systems I: Regular Papers</i> , vol. 65, no. 5, pp. 1591-1601.	2018	SCI
3	S. W. Hussain, T. V. Mahendra, S. Mishra, and A. Dandapat, “Match-line division and control to reduce power dissipation in content addressable memory”, <i>IEEE Transactions on Consumer Electronics</i> , vol. 64, no. 3, pp. 301-309.	2018	SCI
4	T. V. Mahendra, S. W. Hussain, S. Mishra, and A. Dandapat, “Precharge free dynamic content addressable memory,” <i>Electronics Letters</i> , vol. 54, no. 9, pp. 556-558.	2018	SCI
5	F. Begum, S. Mishra, and A. Dandapat, “Low power 10-bit flash ADC with divide and collate subranging conversion scheme”, <i>Scientia Iranica, Transactions D: Computer Science and Engineering and Electrical Engineering</i> , Accepted, pp. 1-20.	2019	SCI

- 6 Sandeep Kumar Singh, Kush Khanna, Ranjan Bose, Bijaya Ketan Panigrahi and Anupam Joshi, “Joint Transformation based Detection of False Data Injection Attacks in Smart Grid”, in IEEE Transactions on Industrial Informatics, vol. 14, no. 1, pp. 89-97 2018 SCI
- 7 Sandeep Kumar Singh, Ranjan Bose and Anupam Joshi, “Entropy Based Electricity Theft Detection in AMI Network”, IET Cyber-Physical Systems: Theory and Applications, vol. 3, iss. 2, pp. 99-105 2018 SCOPUS
- 8 Arun Kumar Sangaiah, Darshan Vishwasrao Medhane, Tao Han, M Shamim Hossain, Ghulam Muhammad, “Enforcing Position-Based Confidentiality with Machine Learning Paradigm through Mobile Edge Computing in Real-Time Industrial Informatics”, IEEE Transactions on Industrial Informatics, February 2019. 2019 SCI, Scopus, IEEE
- 9 Medhane Darshan Vishwasrao, and Arun Kumar Sangaiah, “PCCA: Position Confidentiality Conserving Algorithm for Content-Protection in e-Governance Services and Applications." IEEE Transactions on Emerging Topics in Computational Intelligence, Vol. 2, No. 3, June 2018. 2018 SCI, IEEE
- 10 Pankaj Sharma, Ritesh Bhardwaj, Amitesh Kumar, Shaibal Mukherjee, *Trap Assisted Charge Multiplication Enhanced Photoresponse of Li-P codoped p-ZnO/n-Si Heterojunction Ultraviolet Photodetectors*, Journal of Physics D: Applied Physics 51, 015103. 2018 SCI
- 11 Ritesh Bhardwaj, Pankaj Sharma, Rohit Singh, Mukul Gupta, and Shaibal Mukherjee, *High Responsivity $Mg_xZn_{1-x}O$ based Ultraviolet Photodetector Fabricated by Dual Ion Beam Sputtering*, IEEE Sensors Journal, 99,1. 2018 SCI
- 12 Vivek Garg, Brajendra S. Sengar, Pankaj Sharma, Amitesh Kumar, Aaryashree, Shailendra Kumar, Shaibal Mukherjee, *Sputter-instigated plasmon-enhanced optical backscattering layer in ultrathin solar cells: Application of GZO in CIGSe material system*, Solar Energy, vol. 174, pp. 35-44, September 4. 2018 SCI

- 13 Amitesh Kumar, Mangal Das, Pankaj Sharma, Vivek Garg, 2019 SCI
Brajendra S. Sengar, Arabinda Barman, Alope Kanjilal,
Abhinav Kranti, and Shaibal Mukherjee, *Resistive switching
in reactive electrode-based memristor: Engineering bulk
defects and interface inhomogeneity through bias
characteristics*, Semiconductor Science and Technology,
34 035014
- 14 Rohit Singh, Md. Arif Khan, Pankaj Sharma, Myo Than Htay, 2018 SCI
Abhinav Kranti, and Shaibal Mukherjee, *Two dimensional
electron gases in MgZnO/ZnO and ZnO/MgZnO/ZnO
heterostructures grown by dual ion beam sputtering*, Journal
of Physics D: Applied Physics, 51, 13LT02.
- 15 Prachin Bhoyar, Parul Sahare, S.B. Dhok, R.B. Deshmukh, 2019 SCI
“Communication Technologies and Security Challenges for
Internet of Things: A Comprehensive Review”, AEU-
International Journal of Electronics and Communications,
Vol. 99, pp. 81-99, Feb. 2019.
- 16 Parul Sahare and Sanjay B. Dhok, “Separation of Handwritten 2018 SCIE
and Machine-Printed Texts from Noisy Documents using
Contourlet Transform”, Arabian Journal for Science and
Engineering, Vol. 43, No. 12, pp. 8159-8177, Dec. 2018.
- 17 Parul Sahare and Sanjay B. Dhok, “Separation of Machine- 2018 SCIE
Printed and Handwritten Texts in Noisy Documents using
Wavelet Transform”, IETE Technical Review, Vol. 36, No. 4,
pp. 341-361, Jun. 2018.
- 18 Parul Sahare and Sanjay B. Dhok, “Robust Character 2018 SCIE
Segmentation and Recognition Schemes for Multilingual
Indian Document Images”, IETE Technical Review, Vol. 36,
No. 2, pp. 209-222, Apr. 2018.
- 19 Singh, Bhupendra, and Upasna Singh. "Program execution analysis 2018 SCI-E
in Windows: A study of data sources, their format and comparison
of forensic capability." *Computers and Security* 74: 94-114. DOI:
<https://doi.org/10.1016/j.cose.2018.01.006>

- 20 R. Kumar and Nagendra Kushwaha “Design of dual band and wideband circularly polarised slot antenna for wireless applications,” International Journal of Electronics Letters, vol 6, no. 4, pp. 377-389, 2018. 2018 Scopus
- 21 P. Pandey, A. Shukla and R. Tiwari, “Three dimensional path planning for unmanned aerial vehicles using glowworm swarm optimization algorithm” International Journal of System Assurance Engineering and Management, 9(4), Springer, 2018, pp.836-852. 2018 ESCI, Scopus
- 22 Faridi, S. Sharma, A. Shukla, R. Tiwari and J. Dhar, “Multi Robot Multi-Target Dynamic Path Planning using Artificial Bee Colony and Evolutionary Programming in Unknown Environment”, Intelligent Service Robotics, 11(2), 171-186, Springer, 2018. 2018 SCI
- 23 Jain, A. Mishra, A. Shukla and R. Tiwari, “A Novel Genetically Optimized Convolutional Neural Network for Traffic Sign Recognition: A New Benchmark on Belgium and Chinese Traffic Sign 2019 SCI
- 24 Ranjith Ravindranathan Nair, Laxmidhar Behera, Swagat Kumar, “Event triggered finite time integral sliding mode controller for consensus-based formation of multi-robot systems with disturbances”, IEEE Transactions on Control Systems Technology, vol. 27, no. 1, Jan. 2019 , pp 39-47 2019 SCI
- 25 Ranjith Ravindranathan Nair, Hamad Karki, Amit Shukla, Laxmidhar Behera, Mo Jamshidi “ Fault tolerant formation control of non-holonomic robots using fast adaptive gain nonsingular terminal sliding mode control”, IEEE Systems Journal, vol. 13, no. 1, Mar. 2019 , pp 1006-1017 2019 SCI

7. Other Activities and Institute Events

7.1 Training and Placement

- **Student Internships in Companies and premiere Institutes:** Second and Third year students have done internships in various companies and premiere institutes.

7.2 Sports Activities

- **SPARDHA 2018:** Our students had participated in IIT BHU's inter college annual sports meet named "SPARDHA 2018", which was held at IIT BHU campus from 26th to 28th October. It was a national level event, 1700 students from 72 colleges were participated. From our institute, twelve students participated in four different events namely, Table Tennis, Chess, Carom and Athletics. In this event, our institute performed well and won four medals (2 Gold and 2 Bronze). For participating in this event, participants received financial support from the institute. The total registration fees were borne by the institute and 50% of travel concession by the Govt. of India. Exposure to this type of competitive environment has ensured a proper platform for students to portray their talents.

List of students participated in different events at IIT (BHU) Sports MEET 2018

Sl. No.	Sports Name	Name of the Players
1.	Carrom	Ranjith M., Kishanlal Mali, Vinayak, Siddhartha Nune
2.	Chess	Shivam Dev Singh, Devasish Mahato, Vinayak, Ranjith M.
3.	Table Tennis	Ashutosh Kumar Verma, Kishanlal Mali, Amit Jangid
4.	Athletics	Piyush Deshmukh, Amit Jangid, Pandu Naik, Abhinav kumar



A Glimpse of Sports fest at IIIT Pune

- Volleyball League:** IIIT Pune organized an Intra-college Volleyball League at IIIT Pune. The students of IIIT Pune showed their energy and enthusiasm during the league. Dr. Bhupendra Singh was the coordinator of the event. The prize was distributed to the winning team at the end of the event.

7.3 Extra-Curricular Activities

- **Independence Day:** IIIT Pune celebrated flag hoisting ceremony on Independence Day with Siddhant College in 2018. The students of IIIT Pune organized cultural events such as dance, patriotic song, recitation and speech as a tribute to the sacrifice of martyrs and freedom fighters of India.
- **Gandhi Jayanti:** IIIT Pune celebrated Gandhi Jayanti for a week and aimed to influence the students and staffs with the philosophy of Gandhi. Various activities such as food activity, group discussion, sightseeing, drama and songs, guest lecture on Gandhian philosophy were carried out by the students and staffs of IIIT Pune. The Cleanliness drive was carried out successfully around the campus following the motto 'Swachhata hi Seva'. Director, Dr. Anupam Shukla and Registrar, Dr. S.N. Sapali, IIIT Pune involved in cleanliness drive. Dr. Nilofar Nadaf, Dr. S. P. Gudi and Ms. Ruksheen Vajifdar were coordinators of the whole event. As per the guidelines from MHRD, on the occasion of Mahatma Gandhi Jayanti (2nd October 2018), the "Swachh Bharat Pledge" was taken by all the faculty members, staff and students at IIIT Pune. Director, Dr. Anupam Shukla enlightened the students and faculties about "Swachh Bharat Mission" and everyone participated in 'swachhta' activities at the institute.



Celebration of Gandhi Jayanti

- **Republic Day:** The 70th Republic Day of India was celebrated with and patriotic feeling at IIIT Pune on 26th January, 2019. The ceremony took place in the presence of Honorable, Director IIIT Pune, faculty, staff members and students. The ceremony commenced with the flag hoisting of the National Flag at Campus. Subsequently group dances, group songs and speeches on republic day have been performed by IIIT Pune students. The program ends with National Anthem and with distribution of snacks.

- **Dr. A. P. J. Abdul Kalam Birth Anniversary:** The 87th birth anniversary of Late. Dr. APJ Abdul Kalam which was celebrated as “Reading Inspiration Day 2018” at IIIT Pune as on 15th October 2018. The key objective behind organizing this program was to inspire students by his distinguished life and to enhance their reading practices. Various activities were conducted and the coordinator of the events was Dr. S. P. Gudi.
- **National Unity Day:** IIIT Pune celebrated the Rashtriya Ekta Diwas–Birth anniversary of Sardar Vallabh Bhai Patel on 31st October 2018 by organizing events like, Pledge taking ceremony, Inter-departmental competitions, Essay Writing and Poem, Speech Contests, Drama, Unity Run etc.
- **Central Vigilance Week:** Vigilance awareness week-2018 was observed at IIIT Pune from 29th October to 3rd November 2018 with the theme “Eradicate Corruption–Build a New India”. The main objective to observe Vigilance Awareness week was to bring social awareness amongst students, faculty and staff members and to make them aware of the obligations of individuals in eradicating corruption. During the Vigilance Awareness Week-2018 various activities were organized starting from pledge taking ceremony to conducting group discussion, essay writing competition and Poster Making Competition among institute students and faculty members.



Celebration of National Unity Day and Central Vigilance Week

- **Hindi Diwas:** Under the directives given by MHRD, Hindi Pakhwada was organized from 1st September 2018 to 14th September 2018 by the Hindi Cell of IIIT Pune. The Pakhwada celebration was named as “Aaroh” which means “one which keeps increasing”. IIIT Pune dedicated this Pakhwada to our beloved former PM and Bharat Ratna Late. Shri. Atal Bihari Vajpayee. Following activities were carried out under “Aaroh” Hindi Pakhwada: Signature of faculty members in Hindi, Photography and Hindi caption competition (Chavi), Hindi Quiz competition (Spardha), Hindi Speech competition (Bhashan), Poster making (Theme–Poems of Shri. Atalji), Sketch making (Theme-Shri. Atalji), Hindi self-composed poem

competition (Swarachit), Cut and paste poster making competition (Bhittpatra), Hindi Treasure Hunt (Paheli), Hindi Nibandh competition (Nibandh), Hindi Vaad-vivad competition. The institute distributed prizes to the students and staffs on the last day based on their positions. Dr. Sandeep Singh was the coordinator of the event.

- **National Education Day:** IIIT Pune celebrated the birth anniversary of Maulana Abdul Kalam Azad as National Education day on 11th November, 2018 by arranging various events like essay competition and slogan competition.
- **National Mathematics Day:** The great Indian Mathematician Shrinivas Ramanujan was born on 22nd December 1887. For his memorable and great contributions towards mathematics, the Government of India declared the birth anniversary of Srinivasa Ramanujan as National Mathematics Day. IIIT Pune organized "National Mathematics Day" on the occasion of 131st birth anniversary of Shrinivas Ramanujan on 22nd December 2018. As a part of the celebration, puzzles and quiz competitions were organized by Dr. Saif Nalband and Dr. Chandrakant Guled.



Celebration of National Mathematics Day

- **Netaji Subhash Chandra Bose Jayanti:** IIIT Pune celebrated Netaji Subhash Chandra Bose Jayanti on 23rd January, 2019 with an aim to remember and salute his sacrifice to achieve the freedom of the nation. Various cultural events were organized by the students. Dr. Sandeep Mishra was the coordinator of the event.
- **Vivekananda Jayanti:** IIIT Pune celebrated the birthday of Swami Vivekananda as youth day. The life, philosophy and speeches of Swami Vivekananda were spread through various events. The students were motivated by inspirational speeches of Swami Vivekananda.
- **Traditional Day:** IIIT Pune organized traditional day on 27th February, 2019. Students celebrated the Traditional Day to promote "Unity in Diversity" which is

based on the concept where the individual or social differences in physical attributes, skin color, castes, creed, cultural and religious practices, etc. are not looked upon as a conflict. Rather, these differences are looked upon as varieties that enrich the society and the nation as a whole. Students wore traditional attire from their home state, denoting specific culture of their choice. The celebration was full of cultural programmes where the numerous students showed their talents on the stage which included ramp walk, dance and songs. Dr. S. P. Gudi and Dr. Nilofar Nadaf were coordinators of the event.



Celebration of Traditional Day

- Ek Bharat Shreshtha Bharat:** IIIT Pune took various initiatives for Ek Bharat Shreshtha Bharat. Various events were conducted with an aim to the ‘Unity in Diversity’ of our Nation and to maintain and strengthen the fabric of traditionally existing emotional bonds between the people of our Country; to promote the spirit of national integration through a deep and structured engagement between all Indian States and Union Territories through a year-long planned engagement between States; to showcase the rich heritage and culture, customs and traditions of either State for enabling people to understand and appreciate the diversity that is India, thus fostering a sense of common identity; to establish long term engagements and to create an environment which promotes learning between States by sharing best practices and experiences. Dr. Nilofar Nadaf, Dr. Sandeep Mishra, Dr. S. P. Gudi and Dr. Ranjith Nair were the coordinators of the event.



A glimpse of Ek Bharat Shreshth Bharat

- **InfInITy 2k18:** InfInITy is IIIT Pune's annual flagship coding contest hosted every year on Codechef. A three hour coding with ACM style scoring. InfInITy started in 2017 and was limited only to IIIT Pune. InfInITy 2k18 was made open to everyone and witnessed more than 2000 submissions all over the globe within 3 hours. The winner was Himanshu Jaju from Google London.
- **Winner in Smart India Hackathon 2019:** A team of six students participated in Smart India Hackathon held on 2nd and 3rd March, 2019 at NIT, Warangal. The team from IIIT Pune secured the first position in the national competition and earned a prize of Rs 1 lakh. Dr. Bhupendra Singh from IIIT Pune and Ms. Yesoda Bhargava (Visiting faculty IIIT Pune / Research scientist Samsung Research Lab) were mentors of the team. The news was covered by multiple newspapers.



Smart India Hackathon Winners with Director, IIIT Pune

- **Co-organizer of ICPC 2018:** The International Collegiate Programming Contest (ICPC) is a multi-tier, team-based, programming competition operating under the auspices of ACM. Participation has grown to several tens of thousands of the finest students and faculty in computing disciplines at almost 3000 universities from over 111 countries on six continents. IIIT Pune organized ICPC on 17th and 18th December, 2018 with its co-organizer ABV-IIITM Gwalior. Fifty-three teams from premiere institutes across the country participated in the competition. This event was graced by the presence of various dignitaries such as Padma Shri Prof. S.G. Dhande, former Director IIT Kanpur and Mr. Arvind Vadnekar (Regional Manager BSNL). This event was mentored by Dr. Anupam Shukla, Director IIIT Pune and coordinated by Dr. Nagendra Kushwaha, Assistant Professor IIIT Pune.



ACM-ICPC 2018 Inauguration Ceremony

- Annual Gathering:** C'est La Viewas a 2-day Annual Gathering (7th –8th April 2018) of the college that includes various events wherein students are given chance to think, step forward and showcase their talent. These events include Drama, variety, street play, a speech organized by debate club, DJ night, traditional day and a lot more. Interdepartmental competitions are also organized. This festival also includes an exhibition by arts and crafts team of IIIT Pune.



Annual Gathering

- Educational Tours / Industrial Visits:** Local study tours were arranged for the students. The objective of an industrial visit was to provide us an insight regarding internal working of the industries. During the academic year 2017-18 the following Educational Tours were arranged:

S. No.	Place of visit	Date	Semester	No. of participants
1	Shri. Sant Tukaram Sugar Factory	02/04/2018	2 nd	84
2	Centre for Development of Advance Computing	09/05/2018	1 st	51



(a) Sant Tukaram Sugar Factory, Mulashi, Pune (b) CDAC, Pune

7.4 Invited Talks and Training Programmes

- Dr. B. K. Murthy, Scientist G, DEIT came for an invited talk on **Big Data Analytics: Case Studies** on 17.10.2018. It was an informative and interesting talk. Students gained knowledge in this field and were very satisfied.
- Dr. Mrinal Pandey, Assistant Professor in Manav Rachna University came for an invited talk on **Machine Learning** on 17.10.2018. She explained from the fundamentals of machine learning to its various applications. She also discussed various research problems where machine learning can be applied.
- Dr. Shekhar Bhansali, Professor Florida International University USA came for an invited talk on **Latest Trends in Nanotechnology** on 01.11.2018. He is the department chair of Computer and Electrical department in the university. The talk was very influential and the students understood the importance and several applications of nanotechnology in Computer Science.



A glimpse of invited talks

- Dr. K. K. Pathak came for an invited talk on **Positive Work Environment- Role of Self** on 05.10.2018. He discussed the positive attitude and energetic mind can determine the growth of an organization. Each person should ensure the growth of the organization.
- Dr. A. K. Tripathy, Dean, IIT-BHU came for an invited talk on **Software: Its Science, Engineering and Technology** on 13.11.2018. He briefly discussed the impact of software in the field of science, engineering and technology in today's world.
- Dr. Swagatam Das, Associate Professor in Electronics department at ISI Kolkata, came in two phases for invited talks at IIIT Pune. He delivered a talk on **Doing quality research and getting published in top tier journals- some tips to achieve the dream** on 29.01.2019. The talk was very informative and helped faculties and students how to write research paper for top-tier journals. He also discussed various top-tier journals in various fields. He presented his second talk on **Introduction to Deep Learning** on 14.02.2019. He also discussed fundamentals of deep learning and its various real world applications.
- Dr. Debasish Ghose, Professor in Aerospace Engineering department at IISc Bangalore, came for an invited talk on **Autonomous UAV Swarms** on 20.12.2018. He discussed various computer algorithms that help UAV swarm during missions.
- Dr. Umesh Ghanekar, Professor in Electronics and Communication engineering department at NIT Kurukshetra, came for an invited talk on **Image Processing** on 20.12.2018. The talk was informative and interesting.
- Dr. Phalguni Gupta, Director NITTTR Kolkata, came for an invited talk on **Advanced Topics in Data Structure** on 11.02.2019. He delivered fundamental and advanced topics of data structure and their various applications in real world.
- Dr. A. K. Singh, Professor in Computer Science and Engineering department at NIT Kurukshetra, was invited for a five days lecture series on **Computer Organization and Architecture** for second year CSE and third year ECE students held on 10.12.2018 to 14.12.2018.
- Dr. B. B. Gupta, Assistant Professor in Computer Science and Engineering department at NIT Kurukshetra, was invited for a five days lecture series on **Information Security** for third year CSE students held on 10.12.2018 to 14.12.2018.

- Dr. L. C. Jain, Professor at University of Technology Sydney, Australia, came for an invited talk on **Intelligent Data Processing and Practical Applications in Security** on 03.02.2019. He discussed about data processing approaches and its various applications in real world such as Defence scenarios.
- Dr. Joydip Dhar, Professor at IIITM Gwalior, was invited for a lecture series on **Data Mining and Warehousing** for third year CSE students held on 27.01.2019 to 31.01.2019. The sessions were interactive and the response of the students was tremendous.
- Mr. Anuj Mehra, senior software engineer from MagicBricks Bangalore, was invited for an invited talk on **Data Science and Industry Applications** on 02.02.2019. He discussed fundamentals of data science and various industrial case studies of data science applications. The students realized the requirement of industries.
- Ms. Yesoda Bhargava, research scientist, Samsung Research Lab, came for a lecture series on Innovation held on 06.10.2018 to 07.10.2018. She conducted almost more than twelve hours session on the topic. The sessions included theoretical and practical aspects of innovations. The response of the students was tremendous.
- Dr. Vipul Arora, Assistant Professor in Electrical Engineering department at IIT Kanpur was invited for a lecture series on **Machine Learning and Its Applications** held on 02.02.2019 and 03.02.2019. He discussed mathematical foundations of machine learning and its applications such as speech recognition, recommendation system etc.
- Dr. Nilesh Ware, Assistant Professor in Technology Management department at DIAT Pune, was invited for a lecture series on **Engineering Ethics and Management** on 02.12.2018.
- Mr. C. P. Rathore, senior software engineer at Oracle Bangalore, was invited for a lecture series on **Software Engineering** for third year CSE students held on 16.02.2019 to 20.02.2019 and 29.03.2019 to 31.03.2019. He delivered a total session of thirty-four hours on the topic. He discussed methodologies and recent trends of software engineering followed by industries.
- Mr. Apritam Chowdhary, a JAVA trainer from Kolkata, was invited for **JAVA training Programme** for second year CSE students. He conducted a ten days training programme at IIIT Pune. Students were benefited by the training programme.

- Mr. Sutirtha Vardhan, a soft skill trainer from Kolkata, was invited for a **Soft Skill Training Programme** for second and third year students. He conducted the sessions for almost ten days. Students realized their deficiencies and improved their skills.
- Mr. Anuj and Mr. Anup from CodeChef, were invited for an invited talk for the students regarding coding proficiency and coding competitions. Students benefitted a lot from this talk.

8. Financial Statistics of the year

Balance Sheet as at 31st March 2019

Particulars	Year 2018-19
<u>Sources of Funds</u>	
Corpus/Capital Fund	7,67,59,739.42
Designated/ Earmarked / Endowment Funds	-
Current Liabilities & Provisions	8,59,00,883.00
Total	16,26,60,622.42
<u>Application of Funds</u>	
Fixed Assets	
Tangible Assets	1,64,59,499.00
Intangible Assets	8,92,126.00
Capital Works-In-Progress	47,50,000.00
Investments From Earmarked / Endowment Funds	
Long Term	-
Short Term	-
Investments - Others	5,39,28,046.00
Current Assets	5,15,43,246.42
Loans, Advances & Deposits	3,50,87,705.00
Total	16,26,60,622.42

Income and Expenditure Account for the year ended 31st March 2019

Particulars	Year 2018-19
<u>Income</u>	
Academic Receipts	4,46,82,236.04
Grants / Subsidies	2,38,00,000.00
Income from investments	43,99,368.00
Interest earned	-
Other Income	49,37,639.11
Prior Period Income	53,60,845.00
Total (A)	8,31,80,088.15
<u>Expenditure</u>	
Staff Payments & Benefits (Establishment expenses)	1,67,38,165.00
Academic Expenses	66,33,193.00
Administrative & General Expenses	1,56,65,181.25
Transportation Expenses	9,94,172.00
Repairs & Maintenance	12,45,479.00
Finance costs	-
Depreciation	42,64,785.00
Other Expenses	-
Prior Period Expenses	-
Total (B)	4,55,40,975.25
-	-
Balance being excess of Income over Expenditure (A-B)	3,76,39,112.90
Transfer to / from Designated Fund	-
Building fund	-
Others (specify)	-
Balance Being Surplus / (Deficit) Carried to Capital Fund	3,76,39,112.90

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